

Learning for Data Science

Course 7: Data Processing and Augmentation

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Université
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Mila

CIFAR

Outline

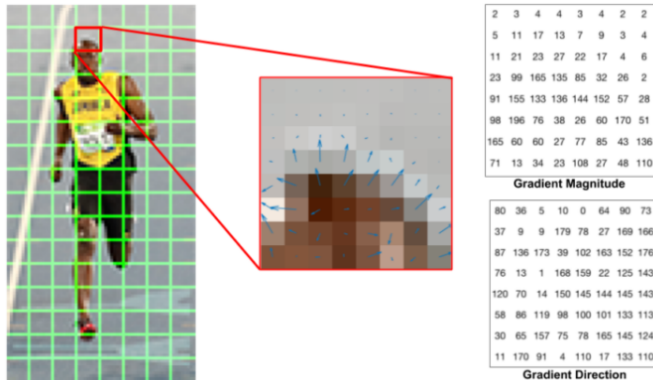
What is a feature?

Motivations for Feature engineering

Data cleaning

Data Transformation

What is a feature?

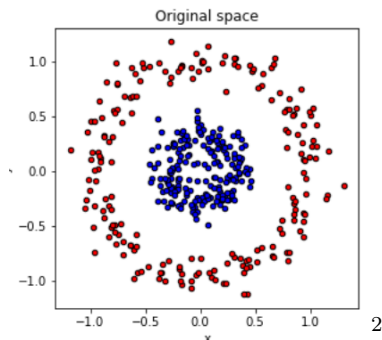


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- Input: x
- Numeric
- Extracted features: $\varphi(x)$

¹source: <https://learnopencv.com/histogram-of-oriented-gradients/>

Motivations for Feature engineering

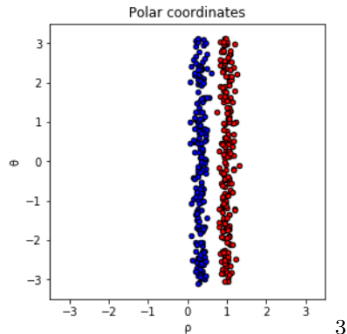
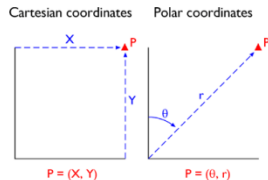
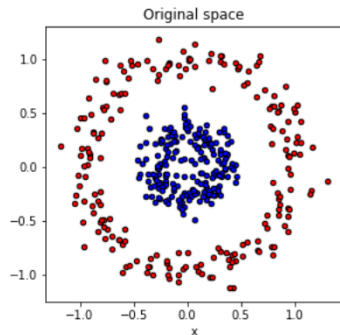


- Make the data separable
- Cannot separate with a linear classifier
 - Use more complex model or another option?

²Source: <https://www.slideshare.net/0xdata/feature-engineering-83511751>

A simple example

- Change of coordinates



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³Source: <https://www.slideshare.net/0xdata/feature-engineering-83511751>

Similar to Prompt Engineering



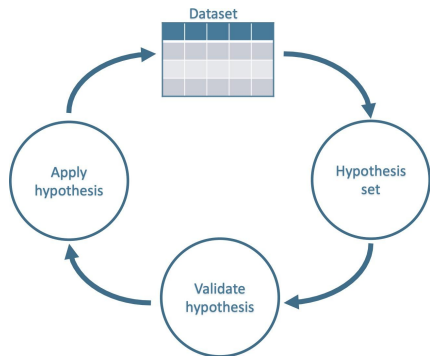
- But, more formal

⁴Source: <https://kripeshadwani.com/what-is-prompt-engineering/>

Some Quotes

- “Coming up with features is difficult, time-consuming, requires expert knowledge. “Applied machine learning” is basically feature engineering.”
Andrew Ng
- “Good data preparation and feature engineering is integral to better prediction”
Marios Michailidis (KazAnova), Kaggle GrandMaster, Kaggle #3, former #1

Feature engineering cycle

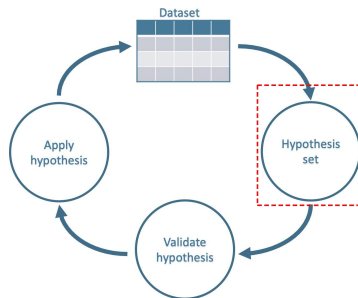


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- Not Feature Engineering
 - Data collection
 - Creating the target variable
 - Removing duplicates
 - Fixing mislabeled classes.

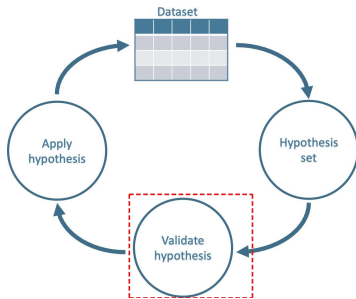
⁵Source: <https://www.slideshare.net/0xdata/feature-engineering-83511751>

Feature engineering cycle



- How?
 - Expert Domain knowledge
 - Prior experience
 - Exploratory Data Analysis (EDA)
 - ML model feedback

Feature engineering cycle



- How?
 - Cross-validation
 - Measurement of metrics
 - Avoid leakage

Feature engineering is hard

- Powerful feature transformations (like target encoding) can introduce leakage when applied wrong
- Usually requires domain knowledge about how features interact with each other
- Time-consuming, need to run thousand of experiments

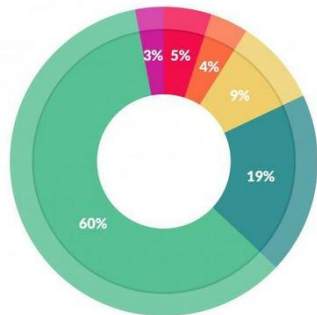
Feature engineering is hard

- Powerful feature transformations (like target encoding) can introduce leakage when applied wrong
- Usually requires domain knowledge about how features interact with each other
- Time-consuming, need to run thousand of experiments
- Why Feature Engineering matters
 - Extract more new gold features, remove irrelevant or noisy features
 - Simpler models with better results
 - Goal: find a global transformation that makes data easier to understand.

Data cleaning



Preparing data takes up a good fraction of a data scientist's time.



What data scientists spend the most time doing

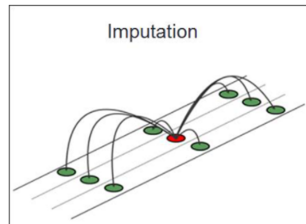
- Building training sets: 3%
- Cleaning and organizing data: 60%
- Collecting data sets; 19%
- Mining data for patterns: 9%
- Refining algorithms: 4%
- Other: 5%

- $60 + 19 + 3 = 82\%$!
- source: <https://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/?sh=40ad122b6f63>

Treatment of missing data

- Frequent problem: Databases with missing data.
- Data imputation (replace missing data)
 - Put generic values (0, Nan)
 - Uses distributions of present x variables to predict missing x variables.
 - Does not work to target y.

- Known data
- Missing data to predict



Example of a database with missing data

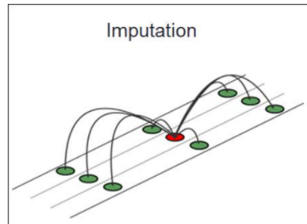
	Caractéristiques (X)					Réponse (Y)
	Surface	Nombre chambres	Quartier résidentiel	...	Nombre écoles	Prix
Maison 1	x_1^1	x_2^1	x_3^1	...	x_{10}^1	250 000
Maison 2	x_1^2	x_2^2	?	...	?	300 000
Maison 3	?	x_2^3	x_3^3	...	x_{10}^3	325 000
...	...	...	...	...	...	...
Maison 100	x_1^{100}	x_2^{100}	x_3^{100}	...	x_{10}^{100}	475 000

- If we remove the three problematic columns, we lose 30% of the data!
- Need to keep data

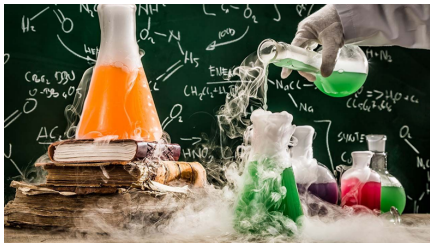
Imputation Methods

- Mean Imputation:
 - Replace with average value
- Regression:
 - Train a model to predict the missing data
- Clustering:
 - Cluster data points together to find class labels
- Substitution:
 - Replace with a new variable unseen in the data

- Known data
- Missing data to predict



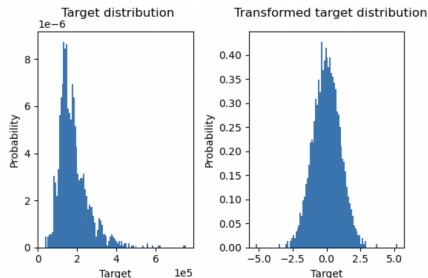
Data Transformation



Target Transformation

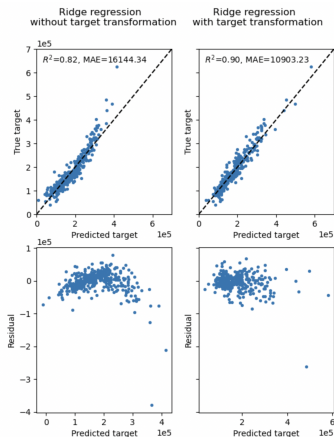
- Transformation of the response variable
- Use it when the variable shows a skewed distribution to make the residuals closer to the “normal distribution” (bell curve)
- Mainly useful for regression tasks: can improve model fit
- for example, $\log(x)$, $\log(x+1)$, $\sqrt{x}$, $\sqrt{x+1}$, etc.
- Reminder: **Many ML training methods assume data is Gaussian, etc.**

Example: Ames housing data



Ames housing data: selling price

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Ames housing data: selling price

⁶Source:

https://scikit-learn.org/stable/auto_examples/compose/plot_transformed_target.html

Characteristic Transformations

- Same idea.

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- In most cases, the order of magnitude of the data changes within the range of the data.

Characteristic Transformations

- Same idea.
- Logarithmic transformation (or log transformation):
- This helps to handle skewed data and after the transformation the distribution becomes closer to normal.
- In most cases, the order of magnitude of the data changes within the range of the data.
- This also decreases the effect of outliers, due to the normalization of magnitude differences.

Transformations

- The data you apply to the log transformation must only have positive values or you will receive an error. Also, you can add 1 to your data before transforming.
- This way you ensure that the output of the transformation is positive.

#Log Transform Example

```
data = pd.DataFrame({'value':[2,45, -23, 85, 28, 2, 35, -12]})
```

```
data['log+1'] = (data['value']+1).transform(np.log)
```

#Negative Values Handling

#Note that the values are different

```
data['log'] = (data['value']-data['value'].min()+1)  
.transform(np.log)
```

	value	log(x+1)	log(x-min(x)+1)
0	2	1.09861	3.25810
1	45	3.82864	4.23411
2	-23	nan	0.00000
3	85	4.45435	4.69135
4	28	3.36730	3.95124
5	2	1.09861	3.25810
6	35	3.58352	4.07754
7	-12	nan	2.48491

Feature coding

- Transform categorical characteristics into numerical characteristics to provide more precise information
- Help to explicitly capture nonlinear relationships and interactions between feature values
- Most machine learning tools accept only numbers as input, e.g. xgboost, gbm, glmnet, libsvm, liblinear, etc.

Converting categorical values to numeric values

- Categorical variables contain categories that are not comparable.
 - Eg: Color = [Red, Green, Blue]
- These variables are converted into binary dummy variables. (Hot Encoding: One-Hot Encoding)
 - Retains all categorical variable information.
 - Suitable for linear models in ML.



- Increases the dimensionality of the variables to process.
- May introduce redundant information.



Each category can be transformed into k dummy binary variables (dummy variables)

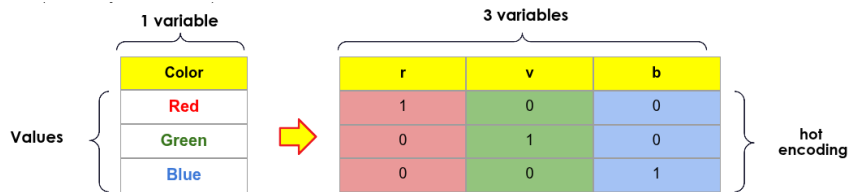


Figure: alt text

- Example: The color blue is coded as $[0, 0, 1]$

Each category can be transformed into k dummy binary variables (dummy variables)

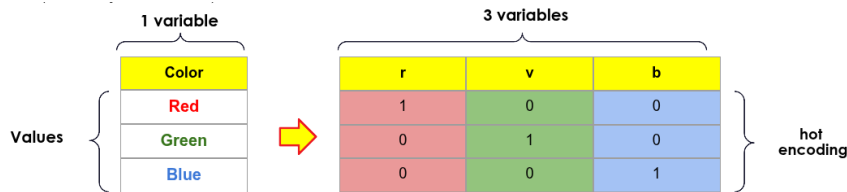


Figure: alt text

- Example: The color blue is coded as $[0, 0, 1]$
- Representation used in
 - Methods based on decision trees
 - Linear regression methods using variable selection

Conversion of ordinal values to numeric values

- Ordinal variables contain categories whose order is important:
 - Eg: `final_grade` = [A, B, C, D, E] with $A > B > C > D > E$
- They are converted into integer values (ordinal encoding)
 - `grade_final` = [0, 1, 2, 3, 4] with $A = 4, B = 3, \dots$
- Representation used in:
 - Methods based on decision trees
 - Linear regression methods if
 - Lots of categories
 - Implicit numeric values of categories are linearly distributed

Frequency encoding

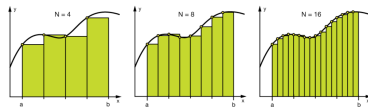
- Encoding of the categorical levels of characteristics in values between 0 and 1 according to their relative frequency

A	0.44 (4 out of 9)
B	0.33 (3 out of 9)
C	0.22 (2 out of 9)

Feature	Encoded Feature
A	0.44
A	0.44
A	0.44
A	0.44
B	0.33
B	0.33
B	0.33
C	0.22
C	0.22

Grouping (Binning)

- Grouping can be applied to both categorical and numeric data:
- Example



Binning illustration of numerical data

#Numerical Binning Example

Value	Bin
0-30	-> Low
31-70	-> Mid
71-100	-> High

#Categorical Binning Example

Value	Bin
Spain	-> Europe
Italy	-> Europe
Chile	-> South America
Brazil	-> South America

Grouping (binning)

- The main motivation for binning is to make the model more robust and avoid overtraining, however, this has a performance cost.
- The trade-off between performance and overfitting is a key point in the clustering process.
- Numerical Grouping: Grouping can be redundant due to its effect on model performance.
- Categorical clustering: Labels at low frequencies probably negatively affect the robustness of statistical models. Thus, the assignment of a general category to these less frequent values helps preserve the model's robustness.

Target average encoding

- It is best to calculate the weighted average of the overall training set average and the level average:

$$\lambda(n) * \text{mean}(\text{level}) + (1 - \lambda(n)) * \text{mean}(\text{dataset})$$

- The weights are based on the frequency of the levels, i.e. if a category appears only a few times in the data set, its coded value will be close to the overall mean instead of the mean of that category. level.
- n : number of examples in the category

Target average encoding

- Instead of encoding categorical variables and increasing the number of features, we can encode each level as the mean of the response.

A	0.75 (3 out of 4)
B	0.66 (2 out of 3)
C	1.00 (2 out of 2)

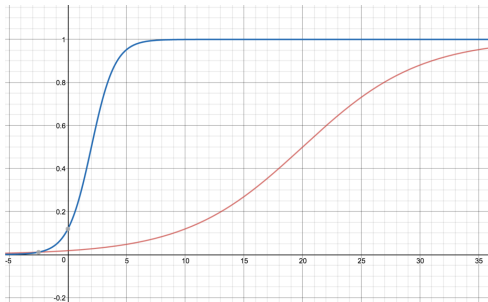
Feature	Outcome	MeanEncode
A	1	0.75
A	0	0.75
A	1	0.75
A	1	0.75
B	1	0.66
B	1	0.66
B	0	0.66
C	1	1.00
C	1	1.00

- Idea: Give the relevant information directly to the model! Problem: can lead to overfitting!!!
- Reminder: A number is needed for training not A, B, C

Target Mean Coding Smoothing Function λ

$$\frac{1}{1 + \exp\left(\frac{-(x-k)}{f}\right)}$$

x = frequency
k = inflection
point
f = steepness



Target Mean Coding Smoothing

$$\lambda = \frac{1}{1 + \exp(-\frac{(x-2)}{0.25})}$$

	x	level	dataset	λ	
A	4	0.75	0.77	0.99	$0.99*0.75 + 0.01*0.77 = 0.7502$
B	3	0.66	0.77	0.98	$0.98*0.66 + 0.02*0.77 = 0.6622$
C	2	1.00	0.77	0.5	$0.5*1.0 + 0.5*0.77 = 0.885$

$$\lambda = \frac{1}{1 + \exp(-\frac{(x-3)}{0.25})}$$

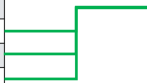
	x	level	dataset	λ	
A	4	0.75	0.77	0.98	$0.98*0.75 + 0.01*0.77 = 0.7427$
B	3	0.66	0.77	0.5	$0.5*0.66 + 0.5*0.77 = 0.715$
C	2	1.00	0.77	0.017	$0.017*1.0 + 0.983*0.77 = 0.773$

Feature	Outcome
A	1
A	0
A	1
A	1
B	1
B	1
B	0
C	1
C	1

Targeted medium encoding: leave-one-out scheme

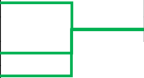
- To avoid overfitting, we could use the leave-one-out scheme

Feature	Outcome	LOOencode
A	1	
A	0	
A	1	
A	1	0.66
B	1	
B	1	
B	0	
C	1	
C	1	



Average target encoding leave-one-out scheme

- To avoid overfitting, we could use the leave-one-out scheme

Feature	Outcome		LOOencode
A	1		0.66
A	0		1.00
A	1		
A	1		
B	1		
B	1		
B	0		
C	1		
C	1		

Average target encoding leave-one-out scheme

- To avoid overfitting, we could use the leave-one-out scheme

Feature	Outcome		LOOencode
A	1		0.66
A	0		1.00
A	1		0.66
B	1		
B	1		
B	0		
C	1		
C	1		

Average target encoding leave-one-out scheme

- To avoid overfitting, we could use the leave-one-out scheme

Feature	Outcome		LOOencode
A	1		0.66
A	0		1.00
A	1		0.66
A	1		0.66
B	1		
B	1		
B	0		
C	1		
C	1		

Average target encoding leave-one-out scheme

- To avoid overfitting, we could use the leave-one-out scheme

Feature	Outcome	LOOencode
A	1	0.66
A	0	1.00
A	1	0.66
A	1	0.66
B	1	0.50
B	1	0.50
B	0	1.00
C	1	1.00
C	1	1.00



Medium target encoding

- When to use it?

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- A target encoding derives numbers for categories using the entity's most important property: its relationship to the target.
- Domain-Driven Features:

Medium target encoding

- When to use it?
- High Cardinality Features:
- Feature with a large number of categories may be difficult to encode: a one-hot encoding would generate too many alternatives
- A target encoding derives numbers for categories using the entity's most important property: its relationship to the target.
- Domain-Driven Features:
- From experience, you might suspect that a categorical feature should be important even if it performed poorly with a metric. Target encoding can help reveal the true informativeness of a feature.

Quiz/Poll

- Why do we need to encode ordinal categories like A, B, C?
- (A) to save memory, (B) because we need numerical values, (C) because they're more human interpretable, or (D) they're faster

Quiz/Poll

- Why do we convert categorical variables like `Color = [Red, Green, Blue]` into one-hot dummy variables?
- (A) to reduce the number of features, (B) to remove redundant information, (C) because models need numerical values and the categories are not comparable, or (D) to make the data easier for humans to read